AP CALCULUS AB

Unit 1 • Lesson 1-1 • Guided notes

Lesson 1-1 • Limits to a Point on a Graph

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Read and write limit notation, including one-sided limits.
- Find limits from a graph and decide whether a limit exists.
- Distinguish the limit of f at c from the value $f(c)$.

Vocabulary. limit, left-hand limit, right-hand limit, open circle, closed (filled) dot, does not exist (DNE), unbounded, oscillating.

1. What a limit means

Key idea

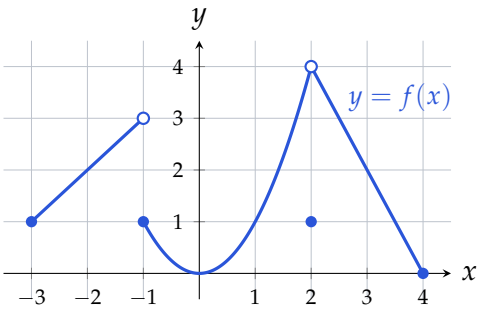
$\lim_{x\to c} f(x) = L$ means: as x gets closer and closer to ____ (from both sides, with $x \neq c$), the values $f(x)$ get closer and closer to ____. The value of f *at* c _____ the limit; $f(c)$ may even be undefined.

2. One-sided limits

$\lim_{x\to c^-} f(x)$ is the _____ limit (uses $x < c$); $\lim_{x\to c^+} f(x)$ is the _____ limit (uses $x > c$).

When does a limit exist?

$\lim_{x\to c} f(x) = L$ exists if and only if $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ both exist and are _____.



Example 1 • Reading the graph of f

Complete the table.

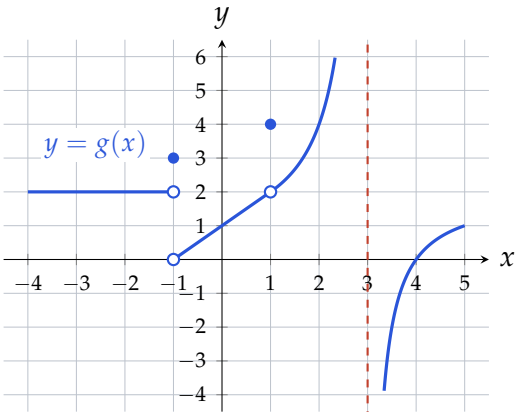
c	$\lim_{x\to c^-} f(x)$	$\lim_{x\to c^+} f(x)$	$\lim_{x\to c} f(x)$	$f(c)$
-1	____	____	_____	____
0	____	____	____	____
2	____	____	____	____
3	____	____	____	____
-3	_____	____	_____	____

3. Three ways a limit fails to exist

Key idea

- **Jump**: the one-sided limits are _____.
- **Unbounded behavior**: $f(x)$ _____ (vertical asymptote).
- **Oscillation**: $f(x)$ keeps _____ between values, like $\sin\frac{1}{x}$ near 0.

Example 2 • A graph with an asymptote



Find $\lim_{x\to -1} g(x)$, $\lim_{x\to 1} g(x)$, $\lim_{x\to 3^-} g(x)$, $\lim_{x\to 3^+} g(x)$, and $\lim_{x\to 0} g\big(g(x)\big)$.

Common mistakes

- Reading the filled dot as the limit.
- Writing "$\lim f(x) = \text{DNE}$". Write "the limit does not exist" and give the reason.

Check your understanding

1. If $\lim_{x\to 4^-} h = 5$, $\lim_{x\to 4^+} h = 5$ and $h(4) = 0$, then $\lim_{x\to 4} h(x) =$ ____.
2. If $\lim_{x\to 1^-} k = 2$ and $\lim_{x\to 1^+} k = -2$, then $\lim_{x\to 1} k(x)$ _____.